

```
# Install (run once)
!pip install torch transformers datasets peft accelerate

# Import libraries
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages (0.18.1)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch)
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Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch)
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Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.3)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from torch)
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Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from torch)
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Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from peft)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from peft)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from peft)

```
model_name = "distilbert-base-uncased"
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
config.json: 100%
```

```
483/483 [00:00<00:00, 12.0kB/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.

```
tokenizer_config.json: 100%
```

```
48.0/48.0 [00:00<00:00, 739B/s]
```

```
vocab.txt: 232k/? [00:00<00:00, 3.11MB/s]
```

```
tokenizer.json: 466k/? [00:00<00:00, 6.18MB/s]
```

```
model.safetensors: 100%
```

```
268M/268M [00:06<00:00, 69.8MB/s]
```

```
Loading weights: 100% 100/100 [00:00<00:00, 119.94it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
```

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status	
-----+-----+		
vocab_layer_norm.weight	UNEXPECTED	
vocab_projector.bias	UNEXPECTED	
vocab_layer_norm.bias	UNEXPECTED	
vocab_transform.weight	UNEXPECTED	
vocab_transform.bias	UNEXPECTED	
classifier.bias	MISSING	
pre_classifier.weight	MISSING	
pre_classifier.bias	MISSING	
classifier.weight	MISSING	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expected it
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training from scratch

```
dataset = load_dataset("csv", data_files="/content/sentiment_dataset.csv")
```

```
def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")
```

```
tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_examples))
test_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_examples))
```

```
Map: 100%
```

```
40/40 [00:00<00:00, 760.80 examples/s]
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)
```

```
peft_model.print_trainable_parameters()
```

WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernel
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925

```
from transformers import TrainingArguments, Trainer
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)
```

```
trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)
```

```
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.
super().__init__(loader)

[5/5 01:25, Epoch 1/1]

Step Training Loss

```
TrainOutput(global_step=5, training_loss=0.693804121017456, metrics={'train_runtime': 125.4387,
'train_samples_per_second': 0.319, 'train_steps_per_second': 0.04, 'total_flos': 5389576273920.0,
'train_loss': 0.693804121017456, 'epoch': 1.0})
```

```
trainer.evaluate(eval_dataset=test_dataset)
```

```
text = "The movie was fantastic and inspiring!"
inputs = tokenizer(text, return_tensors="pt")
```

```
outputs = peft_model(**inputs)
prediction = torch.argmax(outputs.logits, dim=1)
```

```
print("Predicted label:", prediction.item())
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.
super().__init__(loader)

[5/5 00:25]

Predicted label: 1

Start coding or [generate](#) with AI.

